

The code provided for “Magnitude Correction Calculation” helps calculate the extinction and corrected magnitude with the Pan-STARRS filter system. The astronomical objects used were the Andromeda Galaxy and TPS1-10AFX Supernova.

Starting from line 12, the numbers provided under ‘R_ps1’ is the **extinction law**, essentially a dictionary of extinction coefficients. The values provided came from Schlafly & Finkbeiner (2011).

Line 25 calculates the total dimming (A) with the following equation:

$$A = R_ps1[band] * ebv_value$$

This equation multiplies the coefficient (R_ps1) by the dust amount (ebv_value). The product is the light dimmed by the dusts in the specific color.

Line 27 performs the “correction” of the magnitude. The higher the magnitude, the fainter the object. Through this reasoning, the extinction is subtracted to find out how bright the galaxy would actually be if there was no reddening by dust.